A BREED OF ITS OWN

With India set to join the exclusive club of nations that deploy fifth generation stealth fighter jets with the indigenous AMCA, here's a look at how its design compares with the four stealth fighters currently in service





LOCKHEED MARTIN F-35 LIGHTNING II (US)

- ***Development cost:*** **$40 billion (₹3.28 lakh crore)**
- ***Entered service in:*** 2015
- ***Features:*** Multirole fighter that integrates advanced stealth qualities with agility. Cutting-edge sensors and avionics (Northrop Grumman AN/APG-81 AESA radar, BAE Systems AN/ASQ-239 Barracuda electronic warfare system, Lockheed Martin AN/AAQ-40 Electro-Optical Targeting System). Top speed of Mach 1.6 (1,930 kmph)
- ***Armament:*** Two Internal weapons bays capable of carrying air-to-surface and air-to-air weapons (JDAM and Paveway laser bombs, AMRAAM and beyond visual range JATM missiles etc.)

ADVANCED MEDIUM COMBAT AIRCRAFT (AMCA, INDIA)

- ***Development cost:*** **₹15,000** crore for 5 prototypes
- ***To enter service from:*** 2035 onwards

FEATURES:

- The fifth gen, deep penetration stealth jet is designed for multi-role missions (air-to-air and air-to-ground strikes). The AMCA has diamond-shaped trapezoidal wings, the tail wing is an all-moving fuselage mounted stabilator; the vertical stabiliser

dustan Aeronautics Limited (HAL), with support from Indian private firms. The ADA has a 10-year roadmap for making the five prototypes and flight testing thereafter. The first prototype itself would take four years to manufacture. If all goes according to plan, the AMCA will start being manufactured from 2035 onwards.

The AMCA is the most significant military project cleared by the Narendra Modi government before the Lok Sabha election. However, it is learnt that there was opposition, including from a section in the armed forces, against the AMCA on account of its high cost. But its developers told the PMO that the project risked losing its relevance if it did not receive the go-ahead now, and that other countries, just like Turkey has, would steal a march on India in developing a stealth jet.

Military aviation experts claim that Turkey started working on its stealth jet in 2010, a year after India announced its own FGFA programme. In 2015, Turkey announced that it would partner with BAE Systems to design the fighter. In contrast, India and Russia (HAL and Russia's Rosoboronexport and Sukhoi) entered into an agreement to jointly develop a fifth generation jet in 2010, but New Delhi pulled out of it in 2018, after it was felt that the Russians were not willing to share much in terms of technology. The decision to go it alone led to a delay in India's stealth development.

The Indian Air Force (IAF) plans to deploy around 120 AMCA stealth fighters (six squadrons) starting from 2035. The first two AMCA squadrons will consist of the Mark I version, and will be powered by the American GE Aerospace's F-414 engines; the remaining four squadrons will have the more advanced Mark II version, with a more powerful, indigenous engine to be developed jointly with foreign collaboration.



"AFTER YEARS OF HARD WORK, INDIA HAS DEVELOPED ITS OWN STEALTH TECHNOLOGY. INDIA'S AMCA IS GOING TO BE A REAL INDIGENOUS STEALTH JET, NOT A COPY OF ANY EXISTING PLATFORM"

DR ASHISH KUMAR GHOSH
Former project director, AMCA

STATE-OF-THE-ART TECHNOLOGY

To be capable of avoiding enemy radars and air defence mechanism, a stealth fighter uses specialised paint coating and radar-absorbing material to reduce radar bounce-off, exhaust nozzles that reduce infrared radiation emitted by the engine, exhaust plume and fuselage,